

MH370: combining impact and search uncertainty

We estimate where MH370 could have struck the ocean by simulating aircraft motion, comparing predicted satellite measurements with observations, and updating candidate probabilities. Satellite timing (BTO) constrains range; frequency shifts (BFO) constrain motion. We first estimate flights through 00:11 UTC, then simulate their continuation to impact. The cruise ensemble began with 1.02 million candidates, allowing up to 16 changes of direction, eight changes of speed and eight changes of altitude. Commanded speeds span Mach 0.73–0.84 (73–84% of sound speed) and altitudes 25,000–43,000 feet. Distributions of manoeuvre times and control modes are modelling choices. For Bayesian prediction and updating, see Davey et al. (2016, Section 3.1). Fuel exhaustion is conditional on 00:15–00:19 UTC; fuel burn uses a public approximation awaiting aircraft-specific validation. Each independent terminal run selects 12,288 weighted states from that pool.

The final log-on produced two transmissions, labelled R600 and R1200. Their BFOs may be unreliable. We compare four cases: neither, R600 only, R1200 only, or both usable without extra oscillator warm-up adjustments. All retain the corrected R600 BTO and an airborne transmitter through 00:19:37.443. Each separately fitted case receives an initial weight of 25%, before drift/search updates. This specifies a mixture of conditional estimates; it does not infer instrument reliability from the BFO fit scores.

Let $q_m(x)$ be the normalized impact density for case m , $a_m = 1/4$ its initial weight, $L_d(x)$ the debris-drift likelihood, and $L_s(x)$ the probability searches would miss wreckage at x . The equations below update both locations and case probabilities. Set $L_d = 1$ for flight-only results and $L_s = 1$ before searches. This follows Bayesian conditioning and model averaging (Hoeting et al., 1999, p.383).

Drift uses nine recovered objects under a declared GLORYS12/WAVERYS current, wave and object-response model. Drift maps condition each case on available model support before assigning the equal weights; omitted support is reported. Search likelihoods preserve footprint holes. ATSB detection values of 50%, 80% and 94% test uncertainty; 94% is an approximate reference derived from ATSB (2017), Figure 73, p.96. An approximate 2018 outline is tested separately at 60%, with shared or independent missed detections. Recent unlocalized coverage is unassessed.

Across the displayed aerodynamic models and numerical repeats, the drift-plus-ATSB94 mixture places 74–83% north of 35°S. The graphics show 50/90/99% probability contours, with 25 km display smoothing. Original weights determine probabilities. This range is sensitivity, not a confidence interval. Resampling copies candidates. The residual weights have the diversity of just 29–73 equally weighted ancestors; their descendants remain dependent. Omitting one original cruise run shifts some cumulative impact probabilities by up to 8 percentage points. Physical calibration and numerical convergence remain incomplete, so precise hotspots are provisional. A separate synthetic cruise-recovery test still fails. Aerodynamic alternatives remain separate. Future calibrated hydroacoustic likelihoods could similarly update locations and case weights; none is applied here.

$$Z_m = \int q_m(x)L_d(x)L_s(x) dx$$

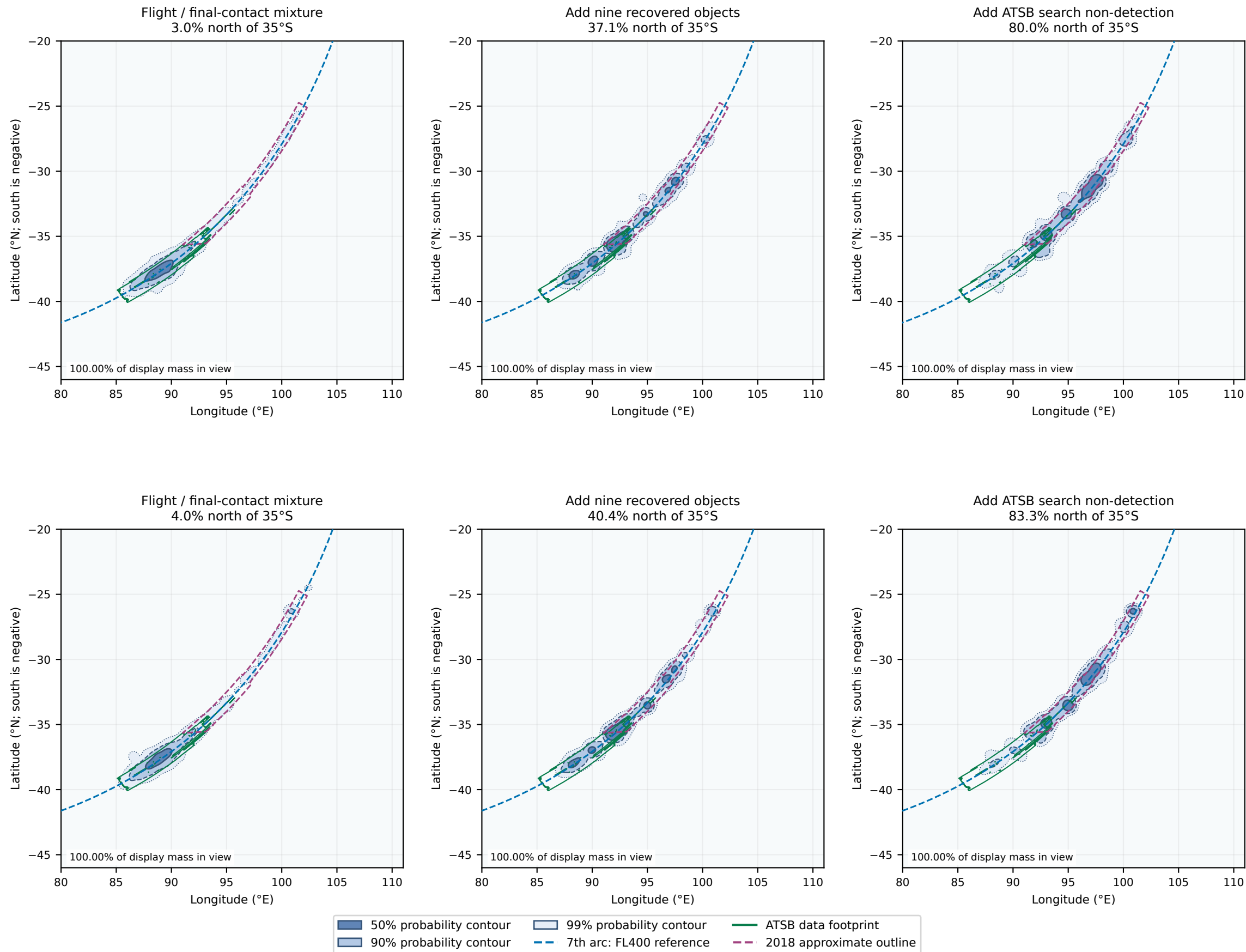
$$a'_m = \frac{a_m Z_m}{\sum_k a_k Z_k}$$

$$p_{\text{res}}(x) = \frac{\sum_m a_m q_m(x)L_d(x)L_s(x)}{\sum_k a_k Z_k}$$

Sources: Davey et al. (2016), Bayesian Methods in the Search for MH370, Section 3.1, pp.13–14.
Hoeting et al. (1999), Statistical Science 14(4), 382–417, p.383.
ATSB (2017), The Operational Search for MH370, Figure 73, p.96. Full source ledger accompanies this report.

How the evidence changes the combined impact distribution

Refined A · Cowling drift repeat A · Top: lower drag; bottom: higher drag



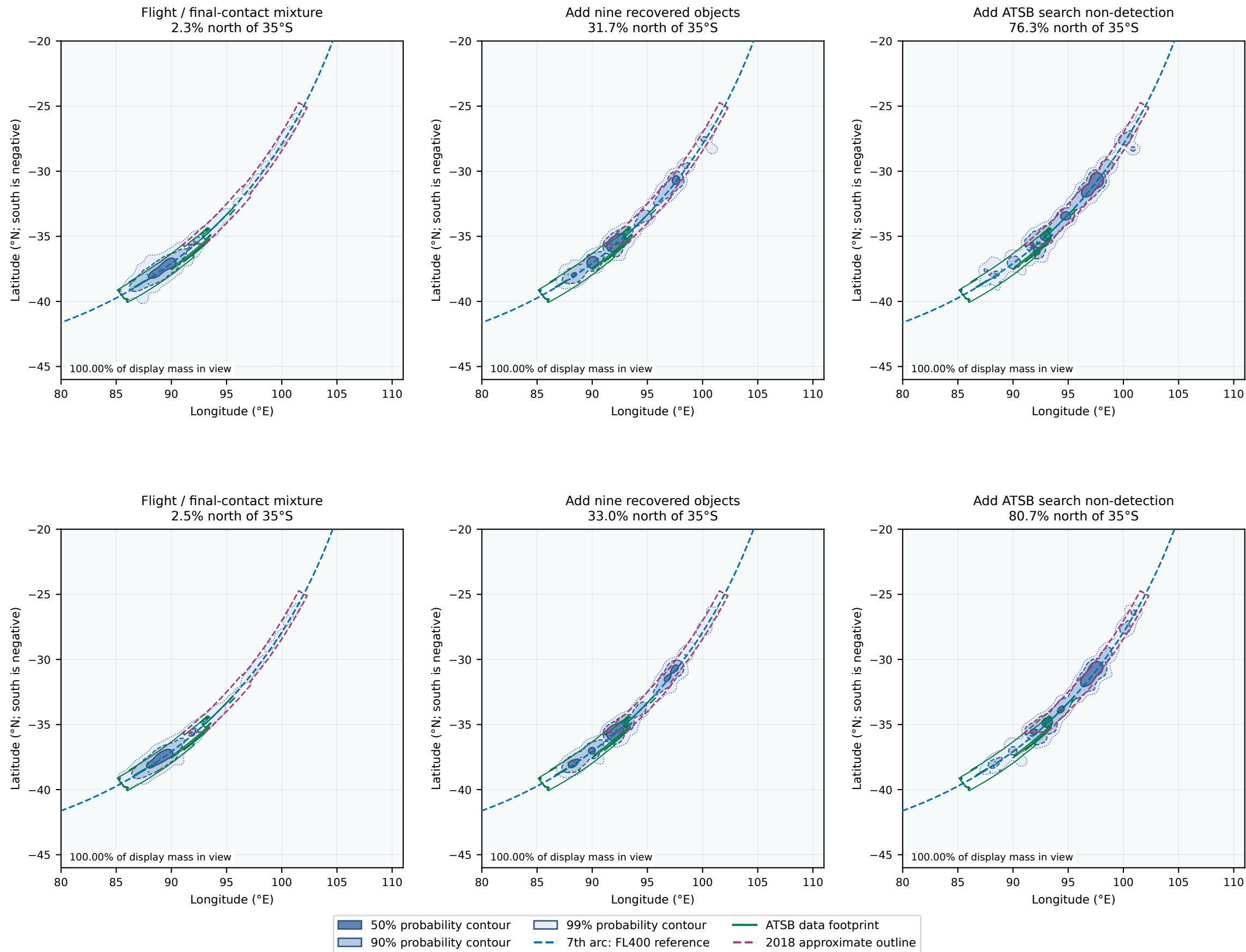
Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.

5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.

Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

How the evidence changes the combined impact distribution

Refined B · Cowling drift repeat A · Top: lower drag; bottom: higher drag



Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.

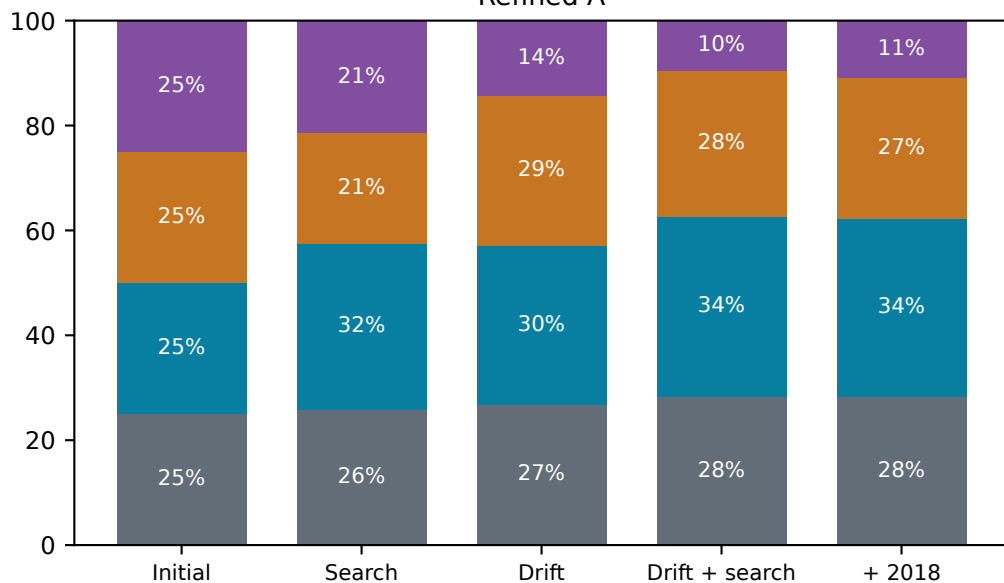
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.

Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

How drift and unsuccessful searches update the four 00:19 cases Equal 25% starting weights after each separate SATCOM fit

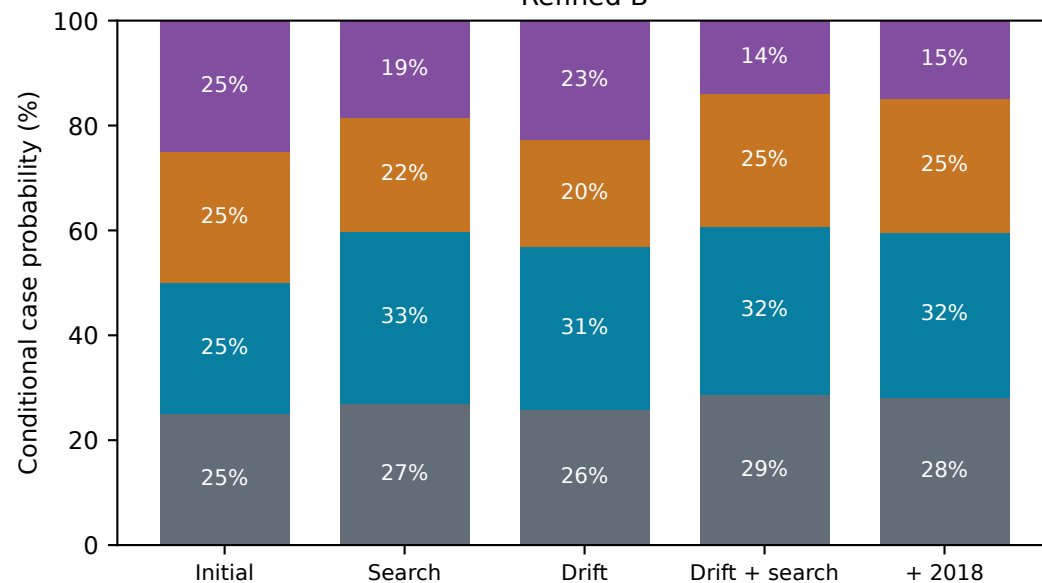
Lower-drag terminal model (OpenAP 26)

Refined A



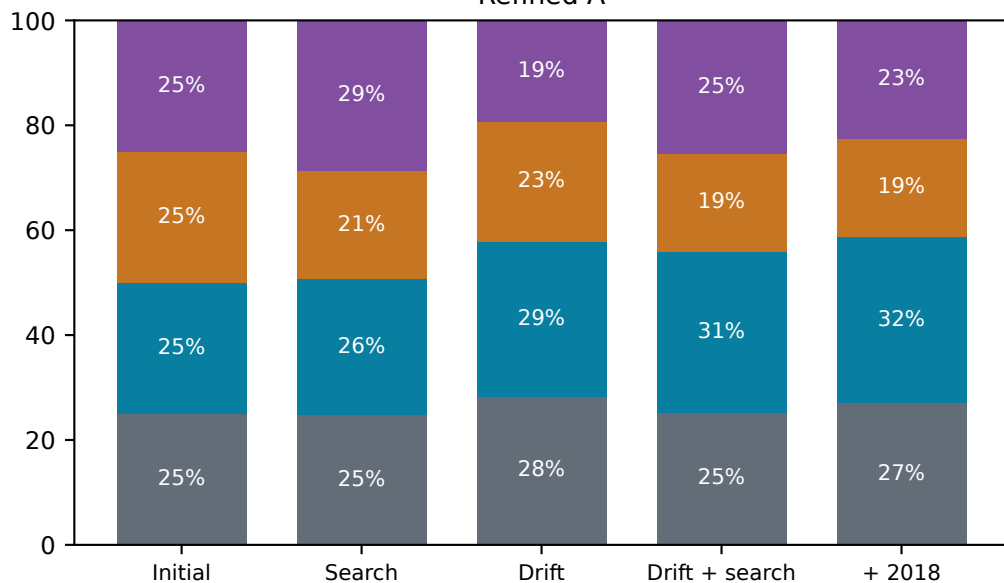
Lower-drag terminal model (OpenAP 26)

Refined B



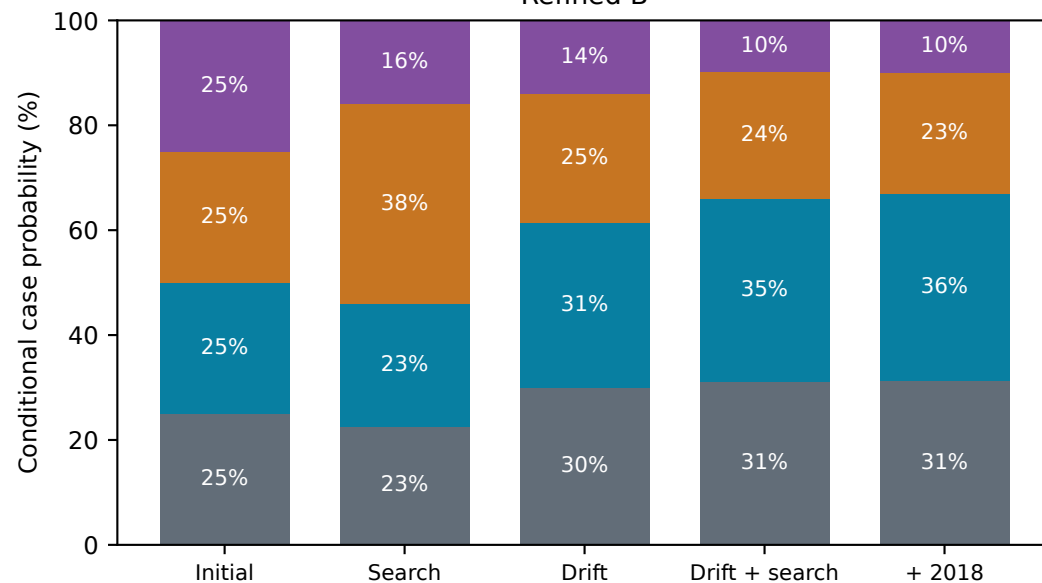
Higher-drag terminal model (OpenAP 2020)

Refined A



Higher-drag terminal model (OpenAP 2020)

Refined B



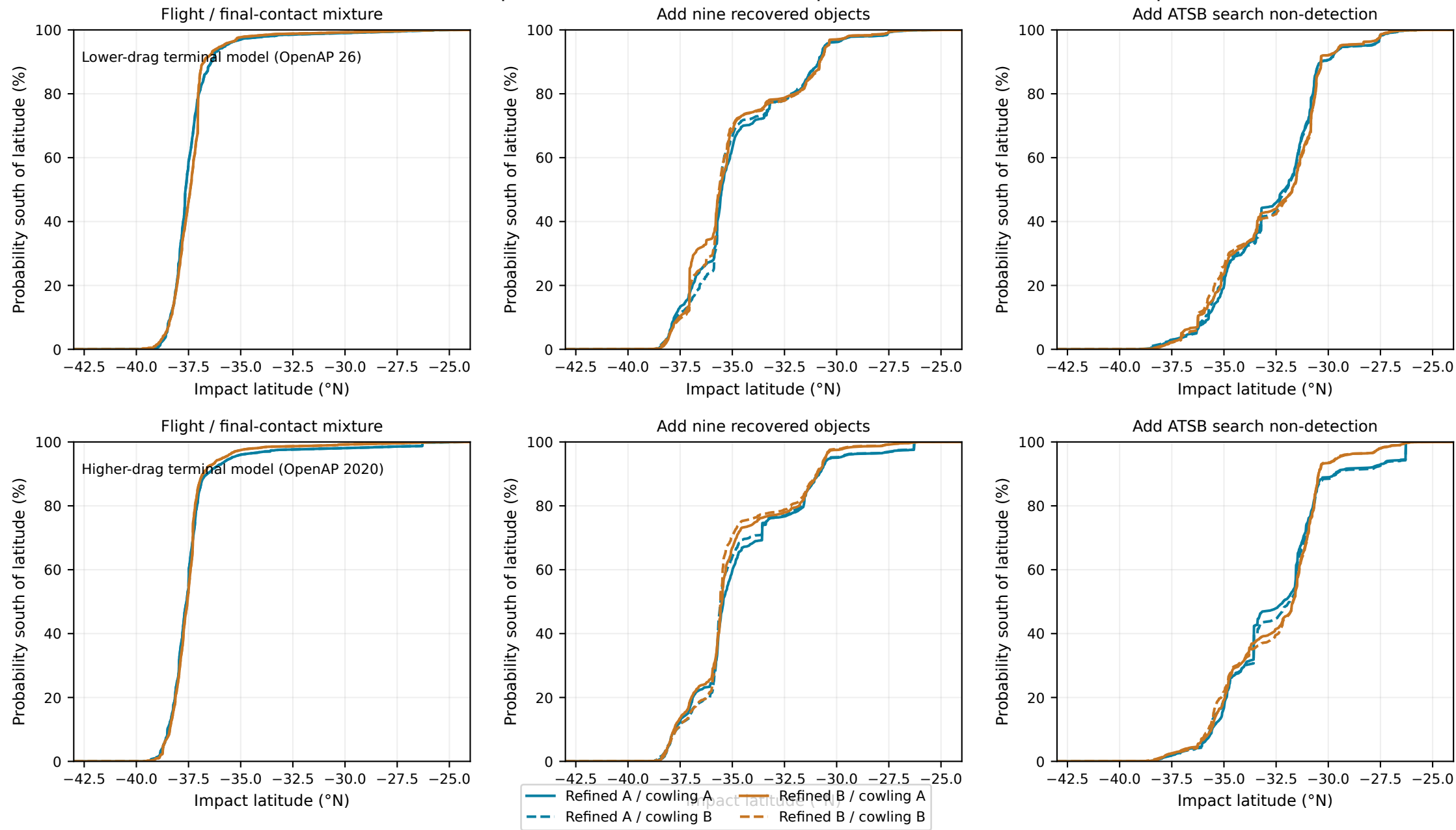
Neither BFO
 R600 BFO
 R1200 BFO
 Both BFOs

Search = ATSB 94% reference. +2018 = approximate outline, 60% effective detection and shared misses.

Cowling estimate A throughout. Different aerodynamic models and independent terminal runs are shown separately; hydroacoustics is absent.

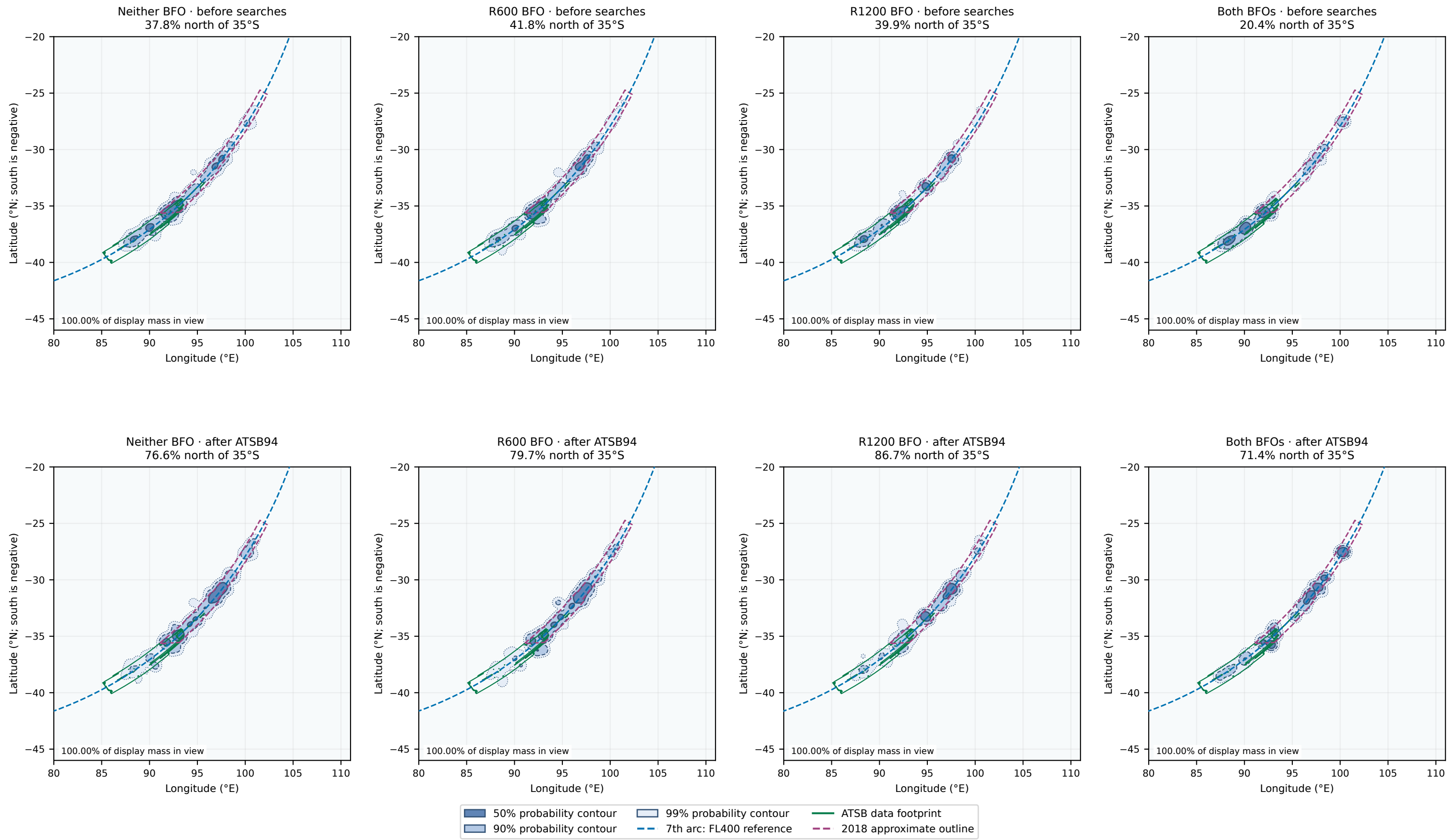
Does the combined impact distribution reproduce?

Unsmoothed cumulative probabilities; the same cruise pool is shared across terminal repeats



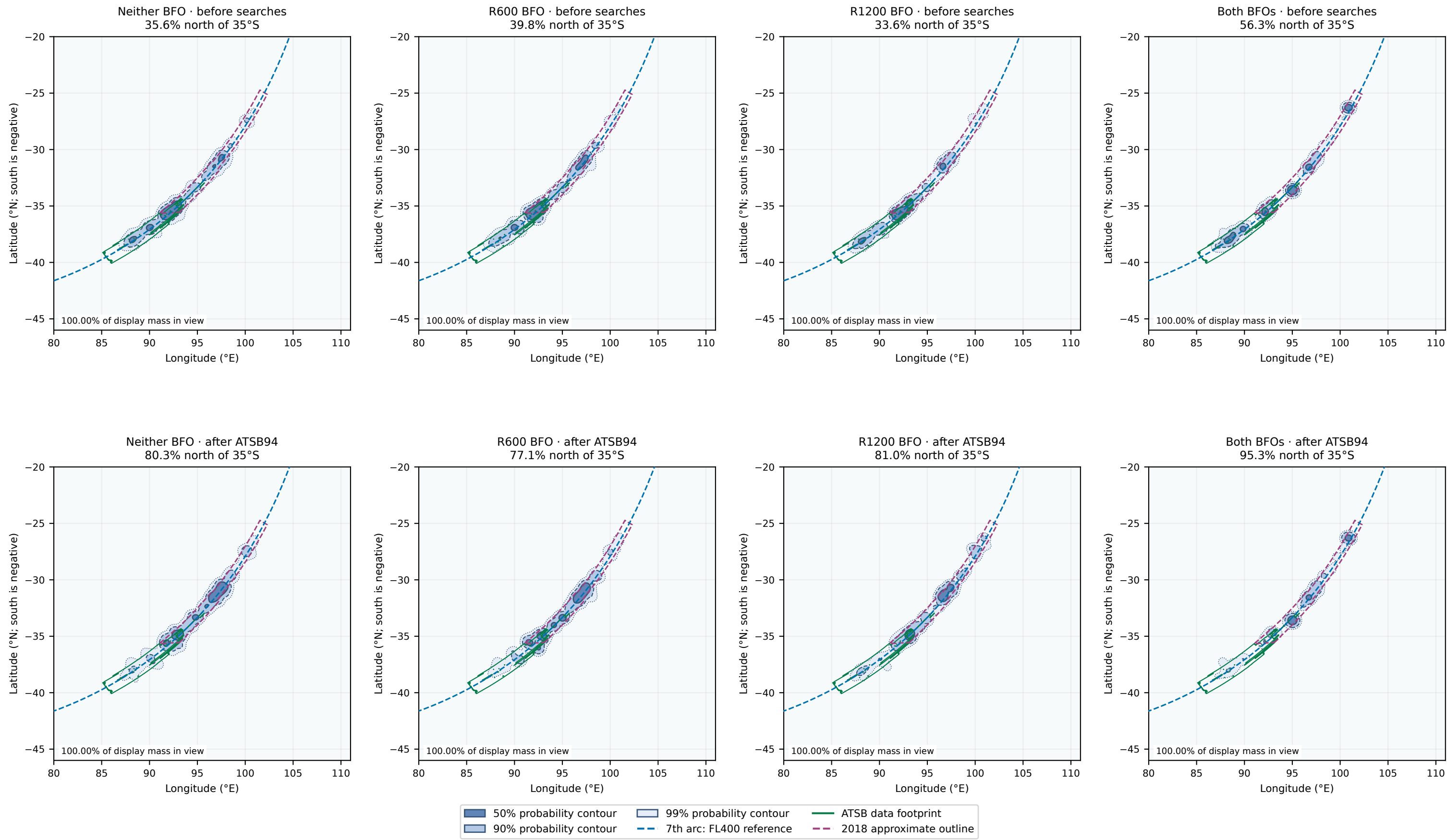
Different physical models are separate rows. Run differences measure numerical variability; these curves are not confidence bands.

Four conditional impact PDFs and their search residuals
Nine recovered objects · Lower-drag terminal model (OpenAP 26) · Refined A · cowlings drift repeat A



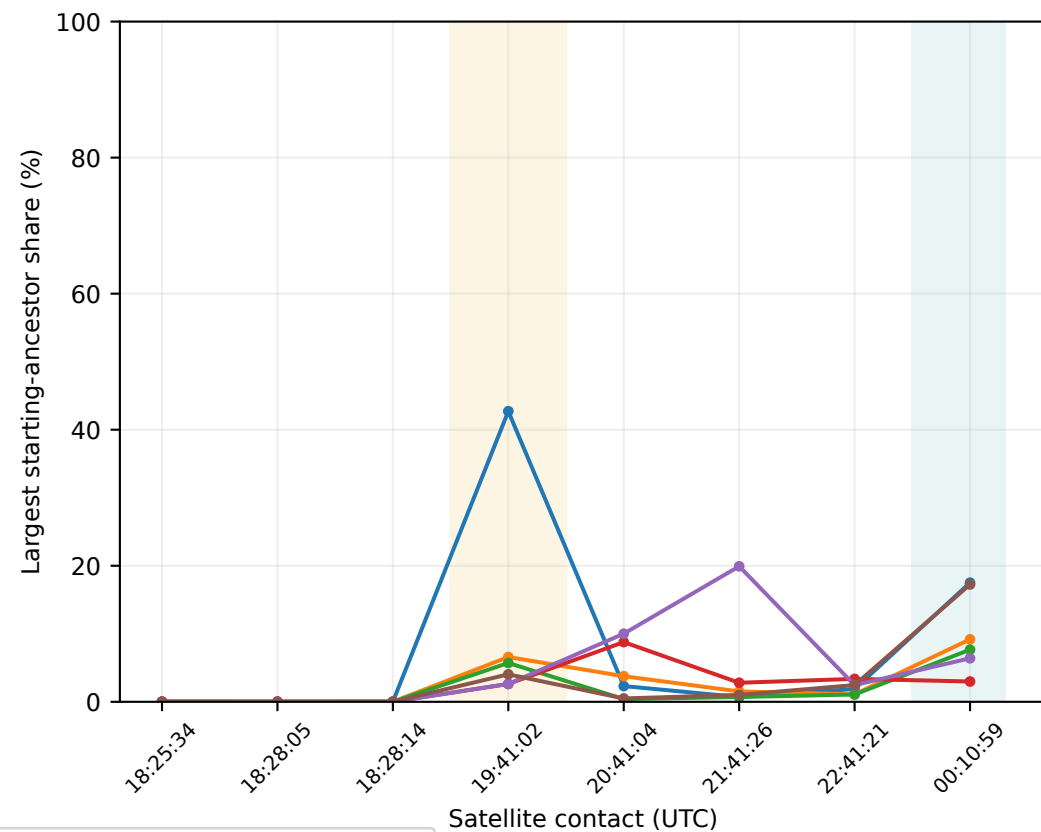
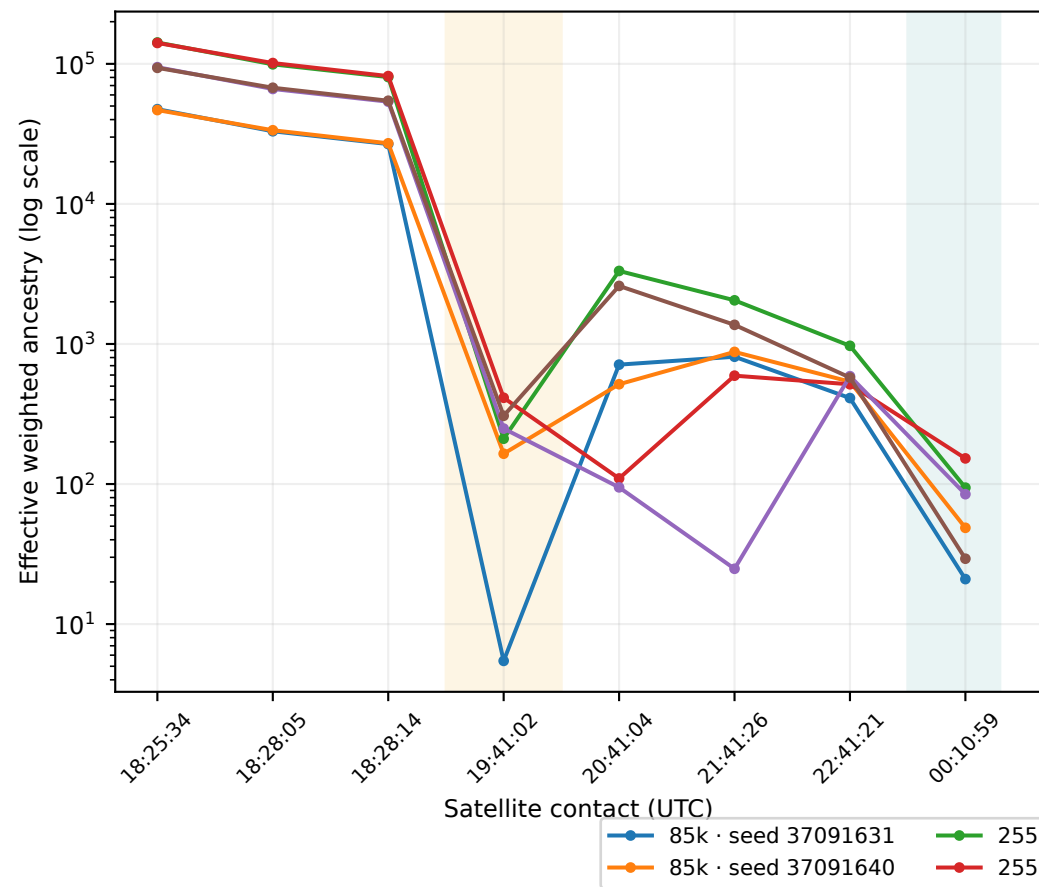
Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.
Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

Four conditional impact PDFs and their search residuals
Nine recovered objects · Higher-drag terminal model (OpenAP 2020) · Refined A · cowlng drift repeat A



Common corrected R600 BTO; airborne through 00:19:37.443 UTC. Fuel exhaustion: 00:15-00:19.
5 km equal-area display grid; Gaussian smoothing $\sigma = 25$ km. Probabilities are calculated from unsmoothed particles.
Conditional on computed impacts; drift panels also condition on native drift support. Numerical and physical validation remain incomplete.

Why a million starting candidates are not a million independent posterior samples

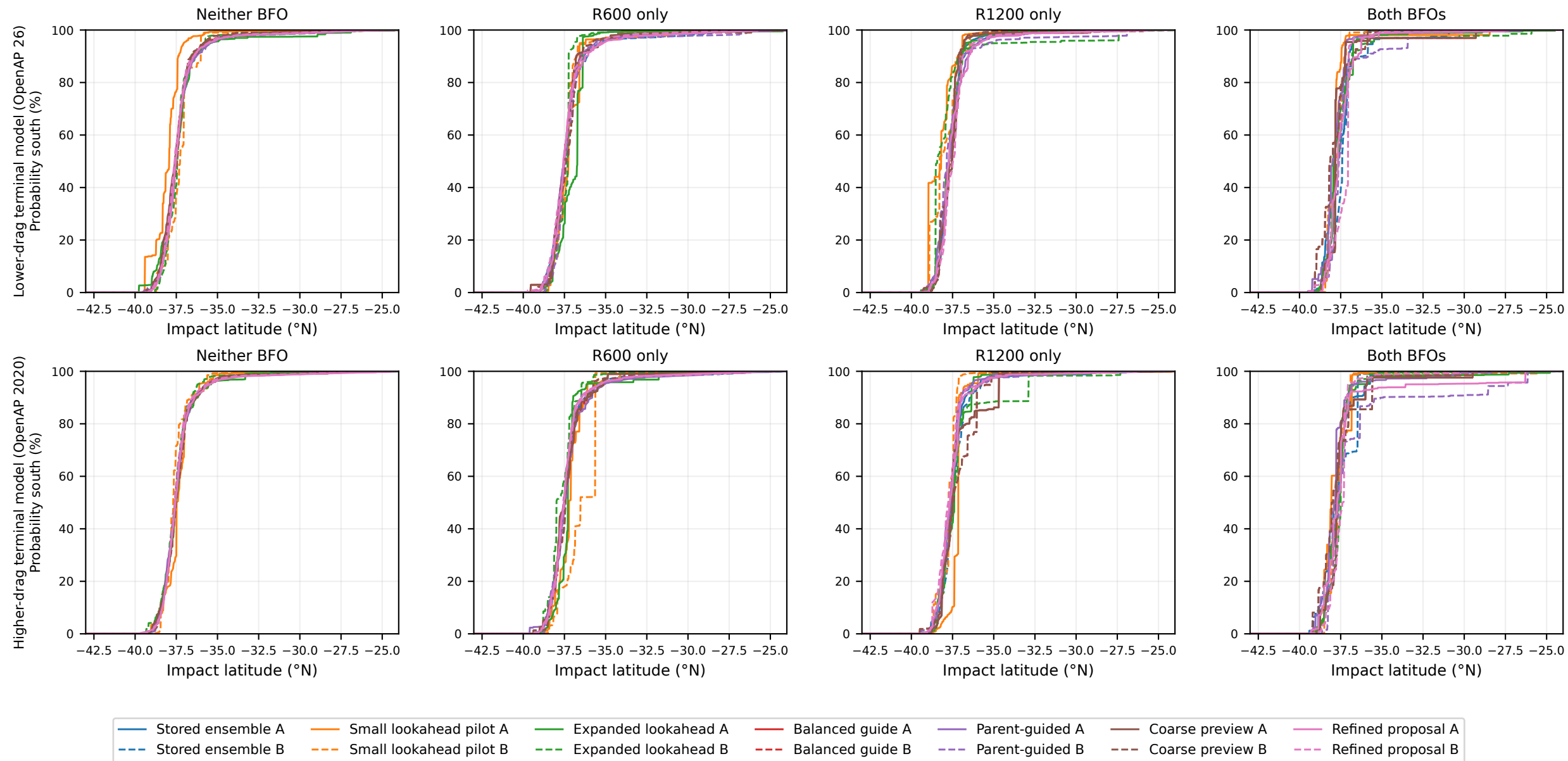


Amber: first long observation gap. Blue: final contact and fuel conditioning. Six original cruise runs; no new simulation.

Effective ancestry = $1/\sum W_r^2$, where W_r sums descendant weights from starting ancestor r .

This diagnoses weight concentration and shared ancestry; it is not a calibrated independent sample count or a convergence threshold.

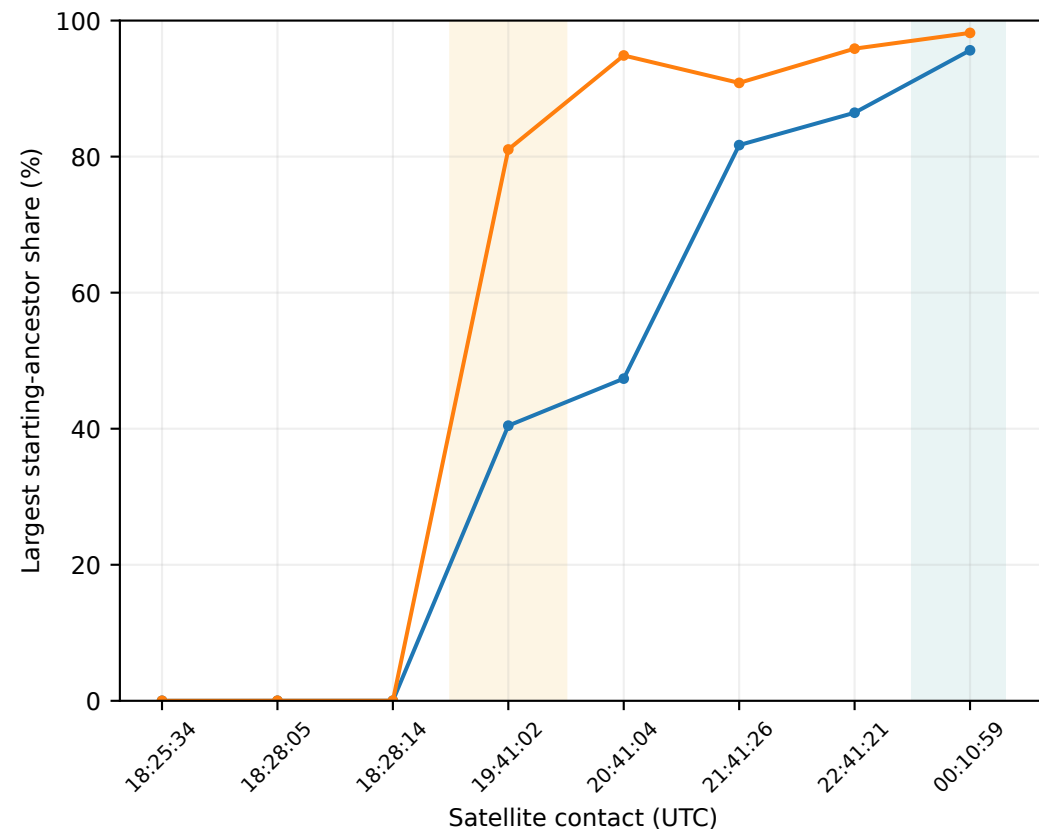
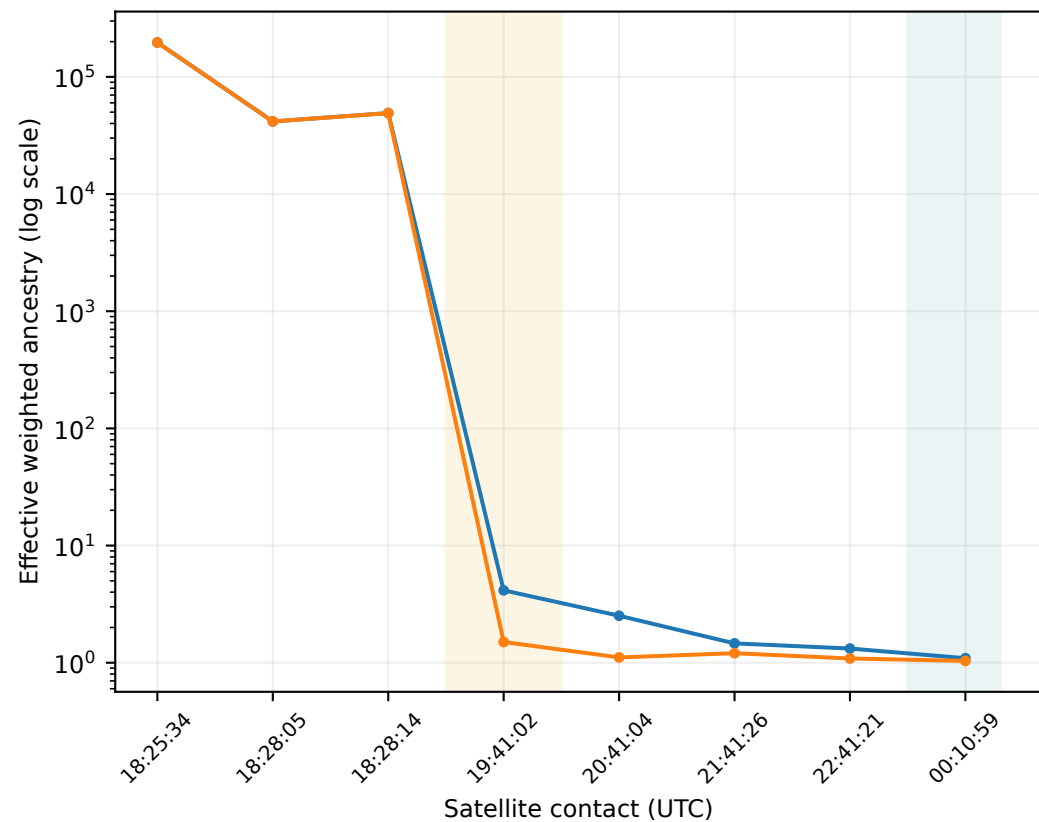
Independent numerical repeats under unchanged physical assumptions



All retain corrected R600 BTO and computed impacts after 00:19:37.443. Drift and searches are absent in this sampler comparison.

Different sample counts and measured costs are recorded in the accompanying table. Pilot agreement alone does not establish accuracy.

Synthetic recovery: concentration develops before the final fuel condition



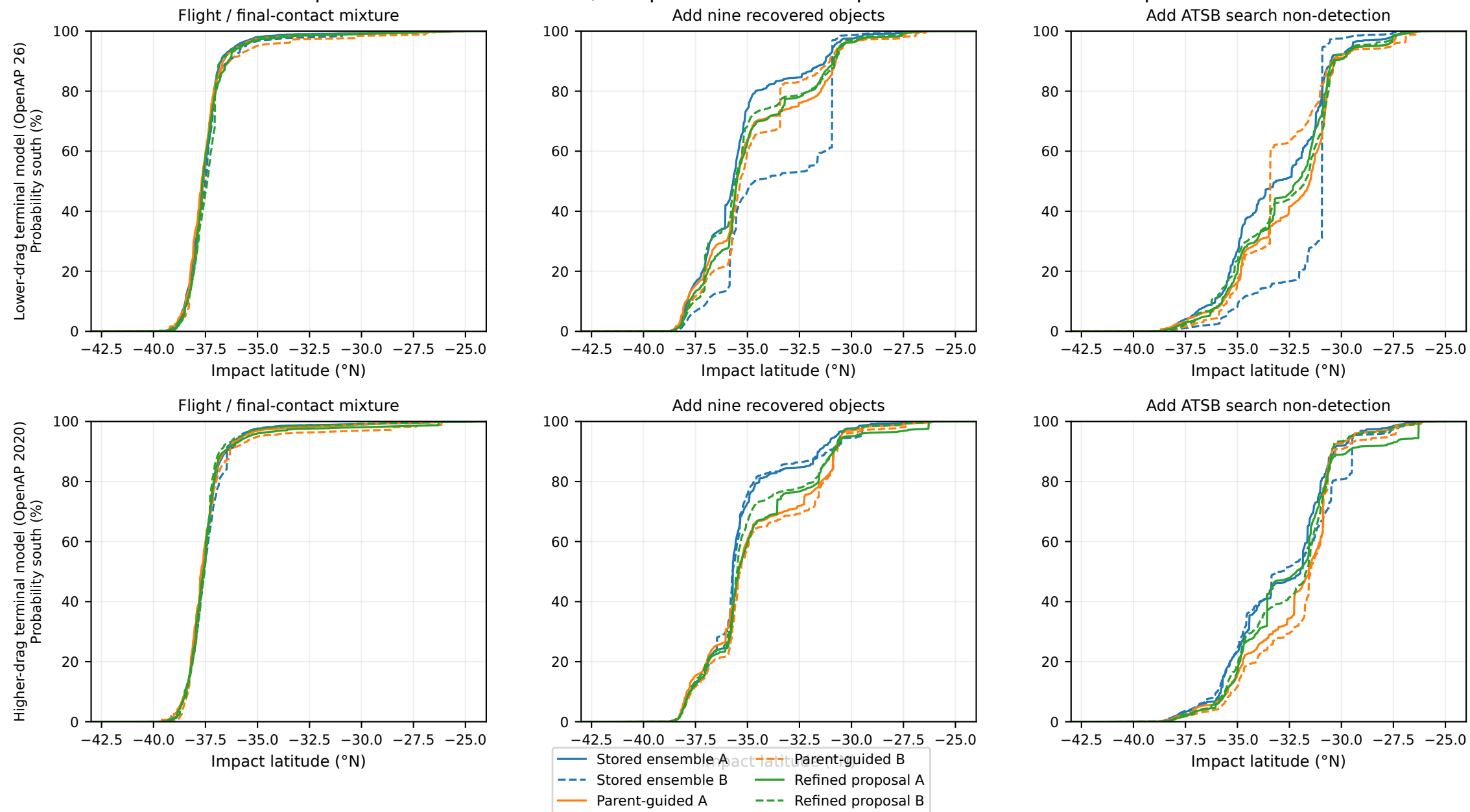
— 500k · seed 37111900 — 500k · seed 37111901

Amber: first long observation gap. Blue: final contact and fuel conditioning. Two saved 500,000-candidate synthetic runs; this is not the MH370 data.

Effective ancestry = $1/\sum W_r^2$, where W_r sums descendant weights from starting ancestor r .

This diagnoses weight concentration and shared ancestry; it is not a calibrated independent sample count or a convergence threshold.

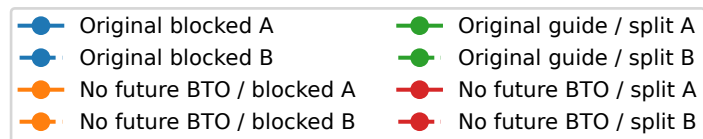
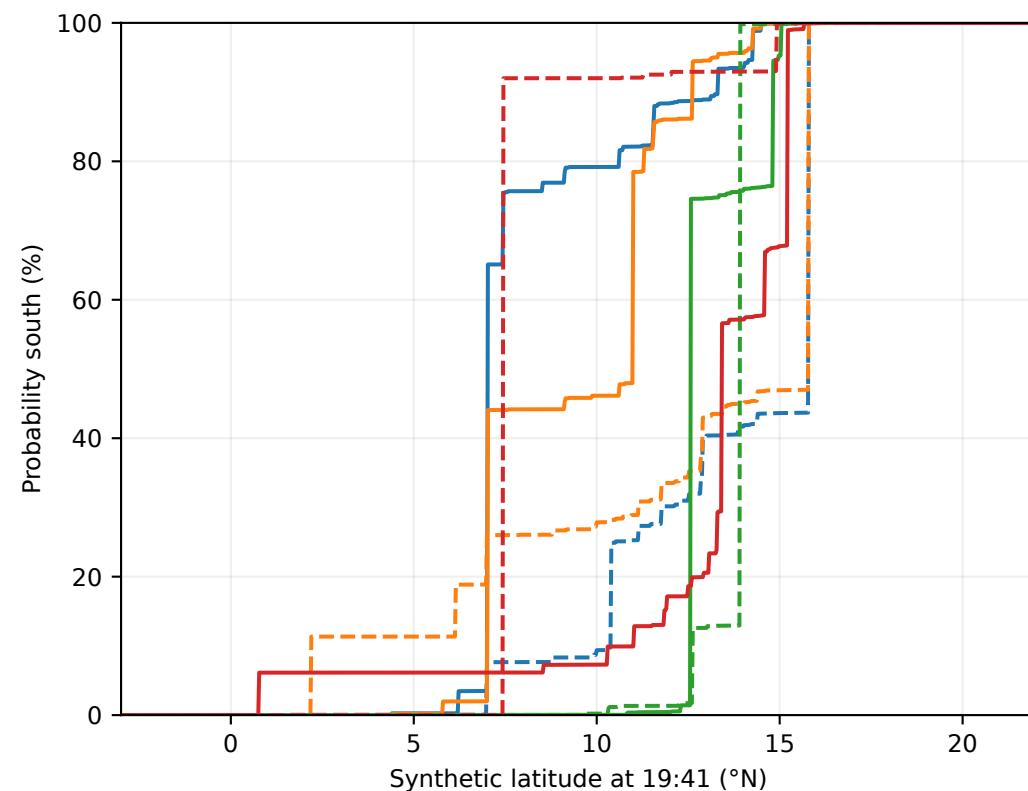
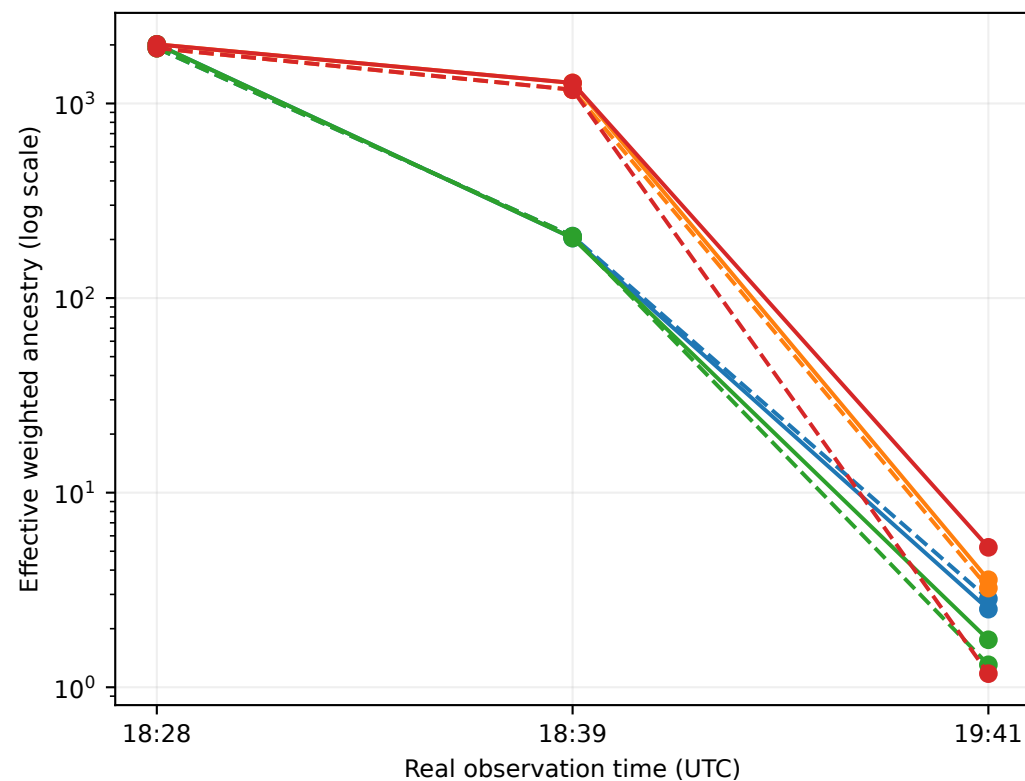
Does improved sampling stabilize the combined evidence model? Equal four-case BFO mixture; independent terminal repeats with the same cruise pool



Cowling estimate A shown. Both cowling repeats and fixed7,500km² cross-run checks are retained in the numerical tables.

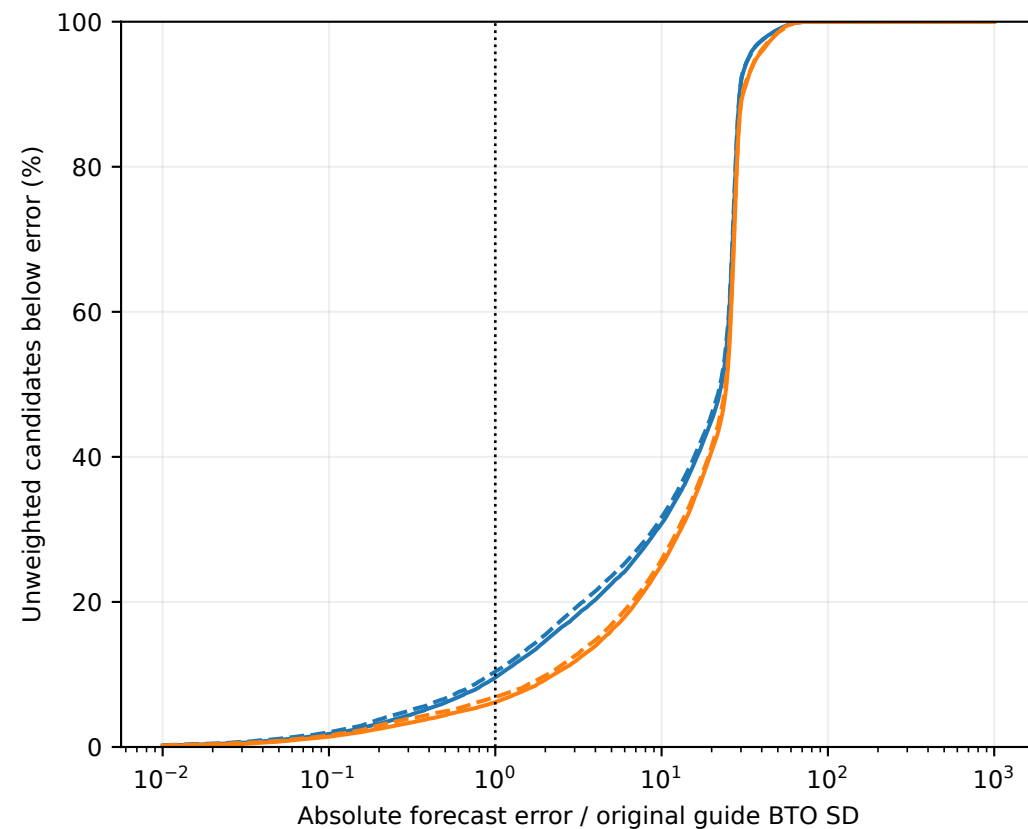
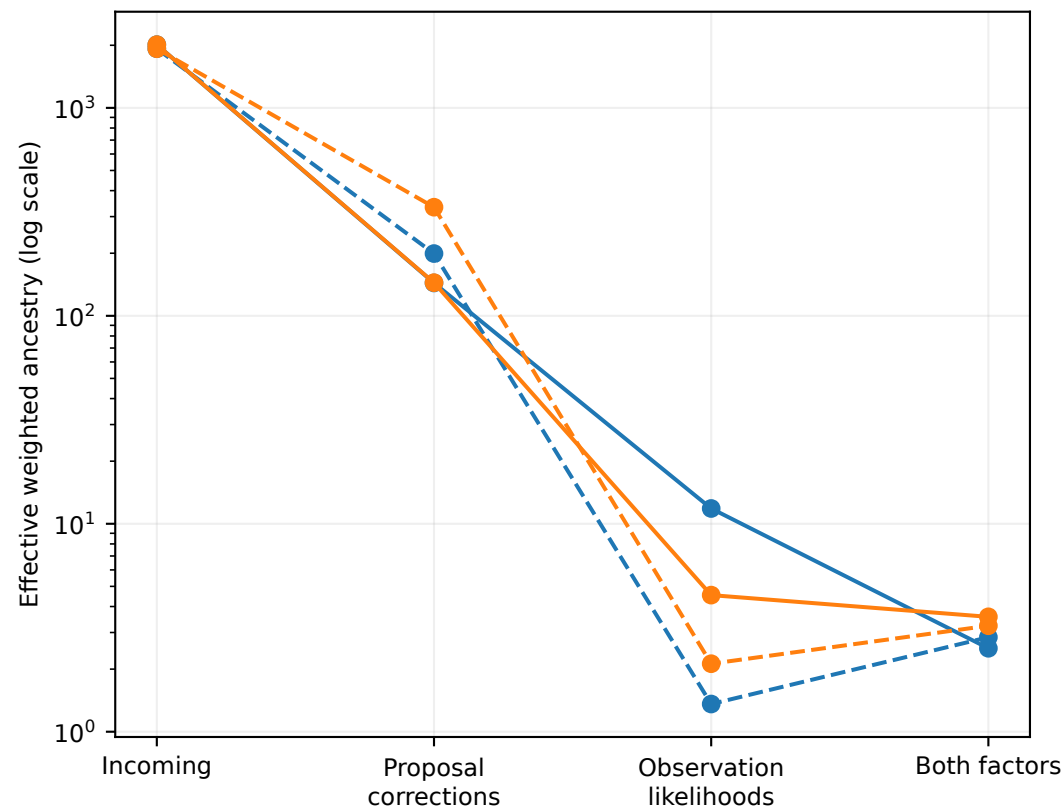
Original probability weights include proposal corrections. Curves show numerical variability; they are not confidence bands.

Cruise-gap controls: changing the guide and the resampling time



Two independent 20,000-candidate seeds per choice; same flight/observation law and verified original 18:28 prefix.
Northern synthetic control data, not MH370 estimates. Fuel conditioning remains at 00:11 and is absent from this diagnostic interval.
A changed guide has exact prior/proposal correction. Resampling at 18:39 adds no measurement or aircraft constraint.

Separating weight concentration from a poor look-ahead approximation



Legend:

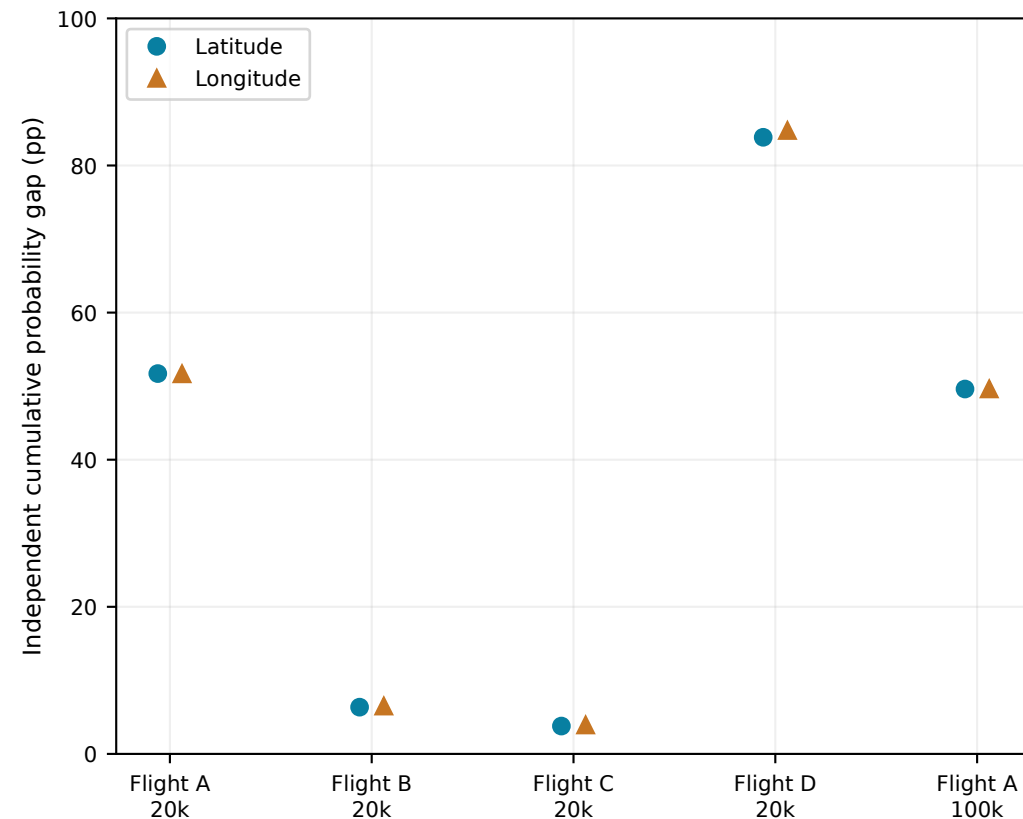
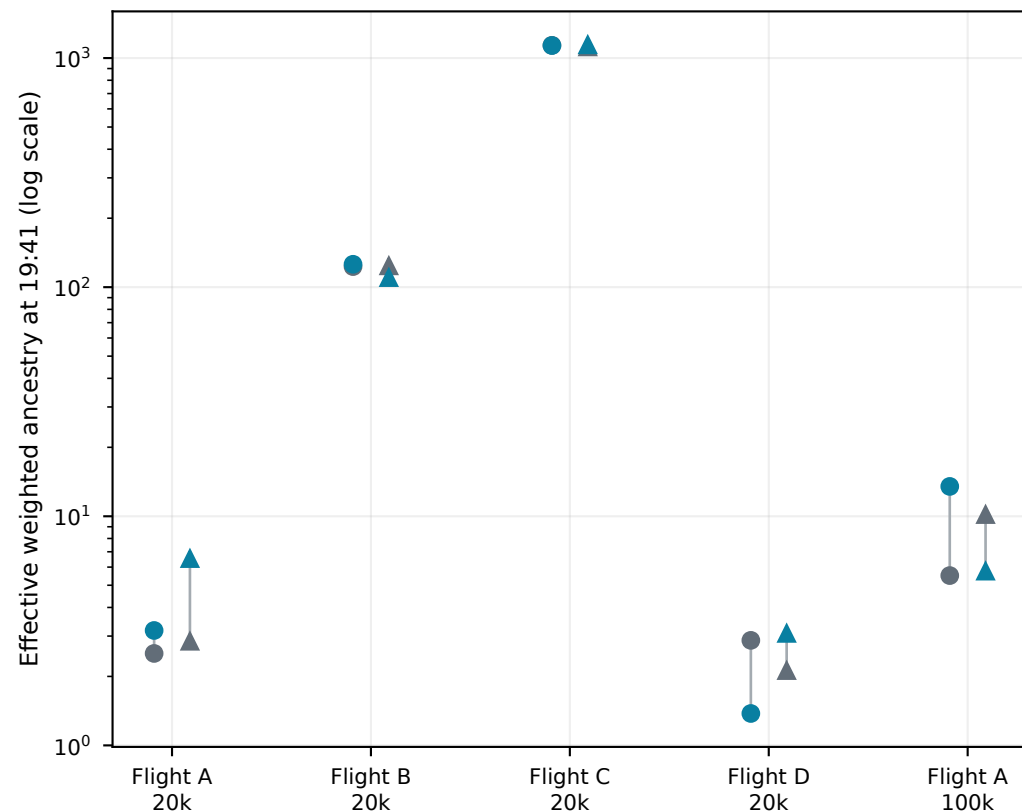
- Original blocked A
- Original blocked B
- No future BTO / blocked A
- No future BTO / blocked B

Left: diagnostic removal of weight terms; the middle columns are not valid alternative posterior distributions.

Right: holding the 18:39 command to 19:41 versus the sampled continuation. Original guide SD: 87 μ s; measurement SD: 29 μ s.

Forecast-error curves describe these sampled proposal banks, not a calibrated future-manoevre distribution.

Joint turn-time guidance does not establish general cruise stability



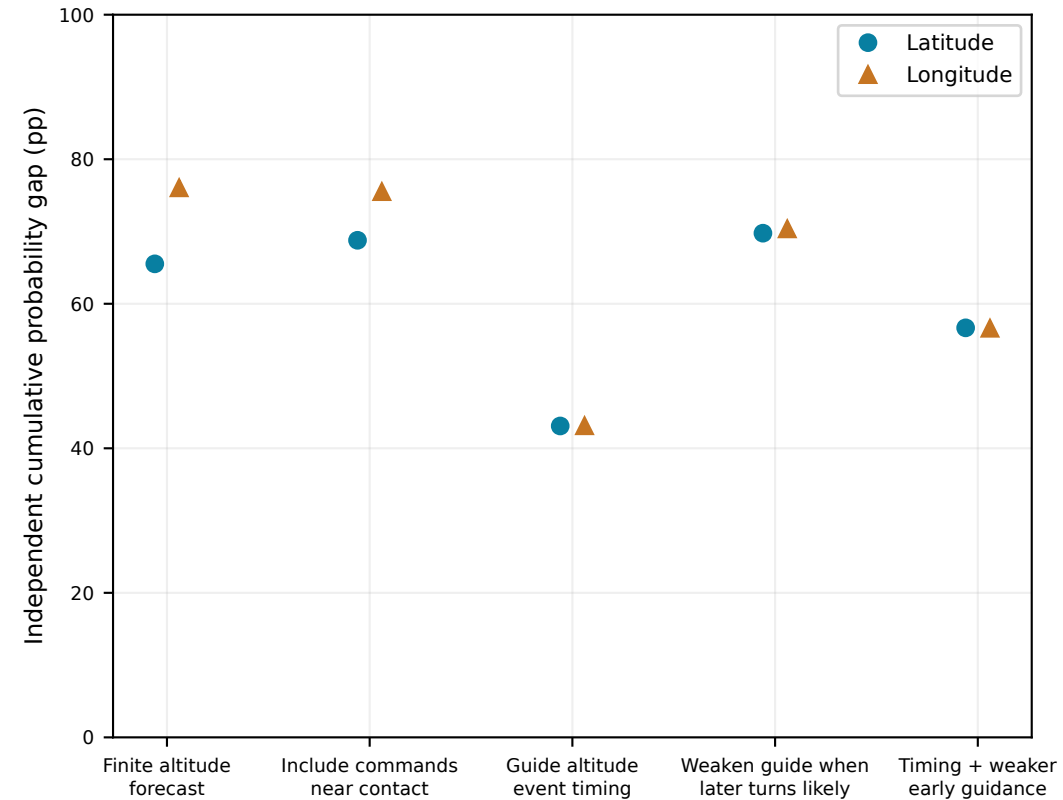
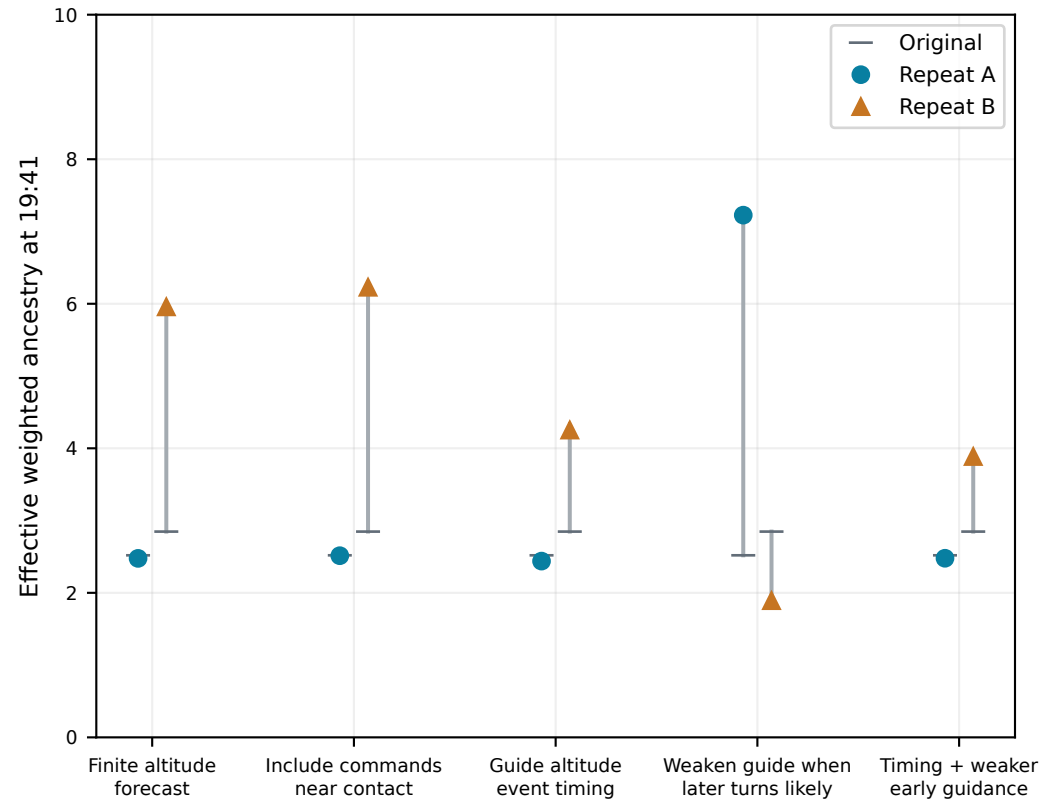
● Original A ▲ Original B
● Joint time/direction A ▲ Joint time/direction B

Four existing synthetic flights; two independent seeds per control. 100k means 100,000 candidates, not independent posterior samples.

Only the first observation gap is tested. The 100k pair took 4.3 minutes; no new full-flight or fuel-conditioned endpoint was simulated.

Same physical priors and exact sampling corrections. This alternative was not integrated; the current estimator default is unchanged.

Altitude and command-guide corrections do not resolve cruise instability



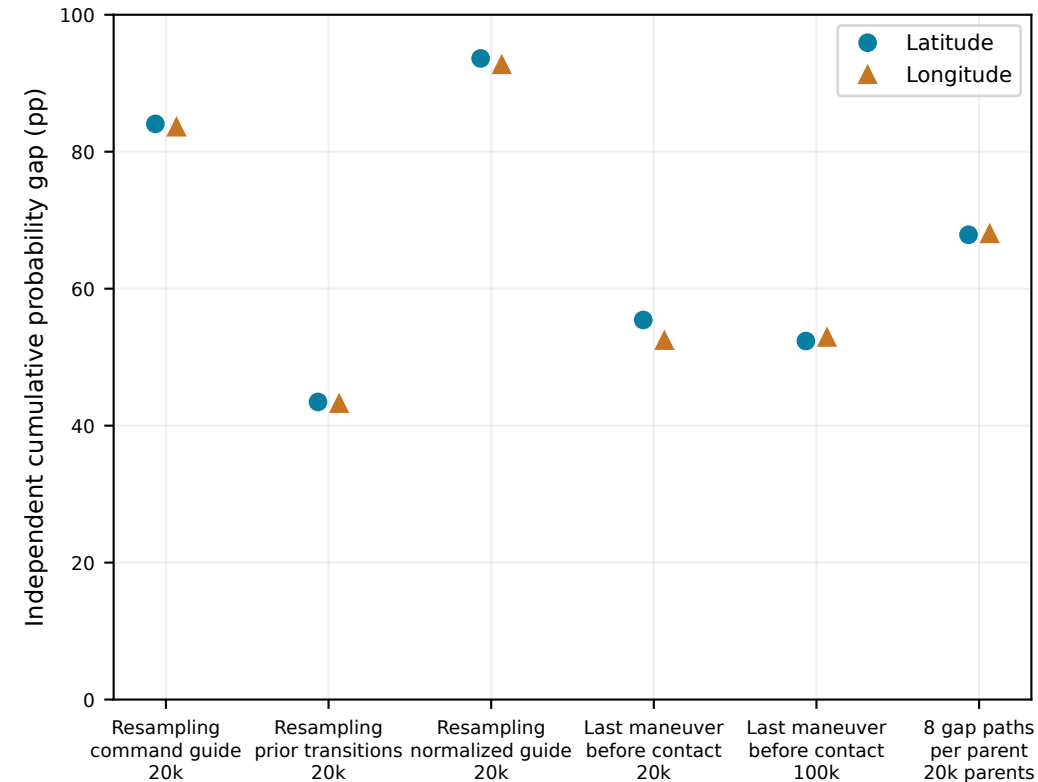
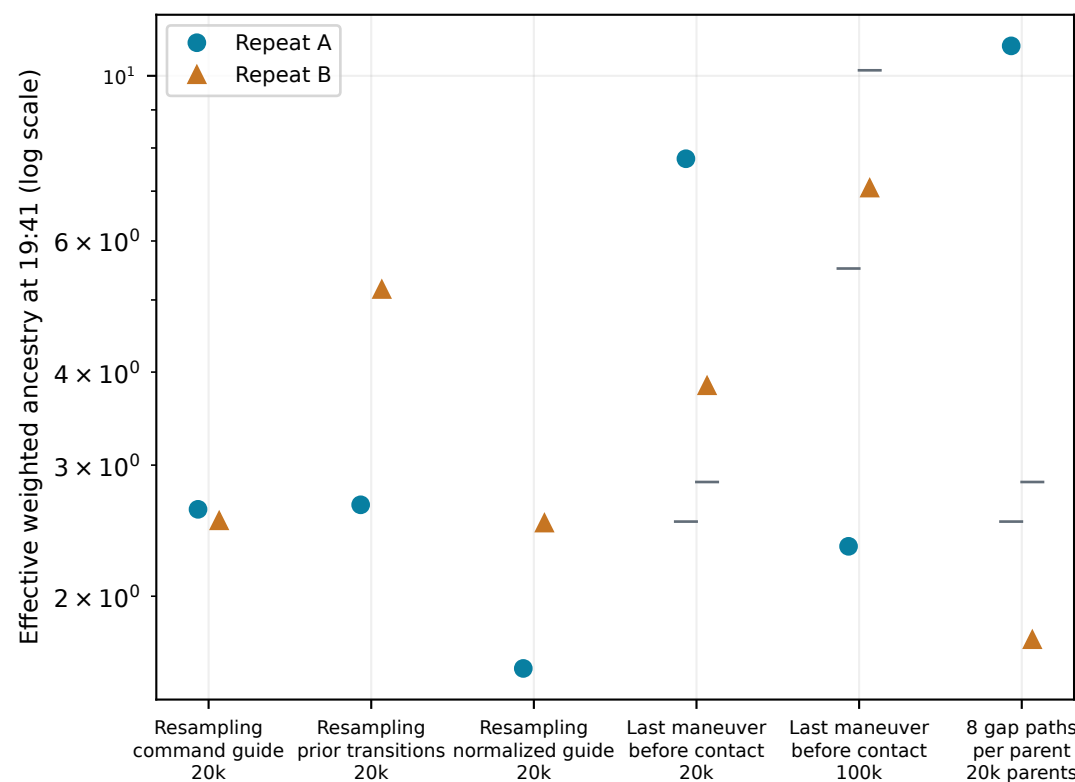
One existing synthetic flight; 20,000 candidates per repeat. Each comparison retains the same original filtering state at 18:28.

Guidance changes only the sampling proposal; exact probability corrections retain the original flight and observation model.

Altitude timing recovers previously poorly sampled climbing paths, but the resulting distributions still depend strongly on the random seed.

These alternatives were not integrated. No full-flight rerun or extra physical evidence is included in this diagnostic.

Cruise interval proposals: correctness and repeatability are separate checks



One synthetic test flight; matched saved starting populations, physical priors and observations. k denotes thousand starting candidates.

Grey ticks show original ancestry. Candidate controls propagate several complete gap paths per parent; these are not independent initial states.

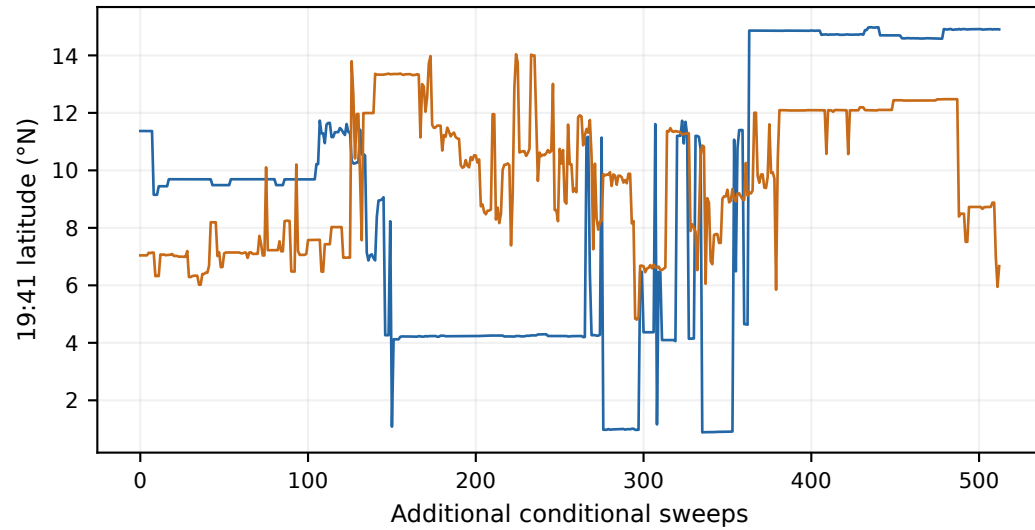
Intermediate potentials cancel; event and candidate proposals retain their exact probability corrections.

These are numerical controls through 19:41, not MH370 location estimates or a demonstration of recovery through 00:11.

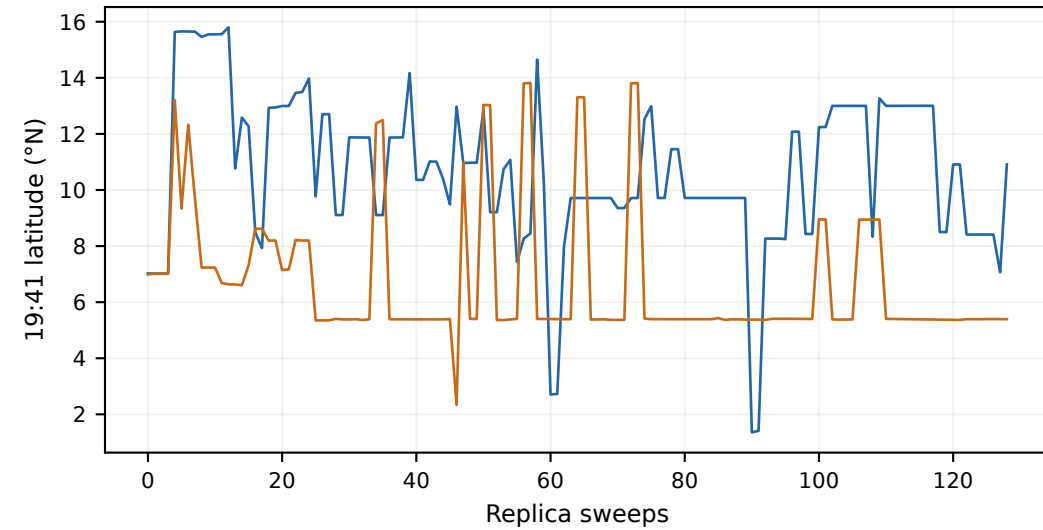
Revising earlier histories: particle Gibbs with full ancestor sampling

Synthetic first-gap control through 19:41; not an MH370 impact map

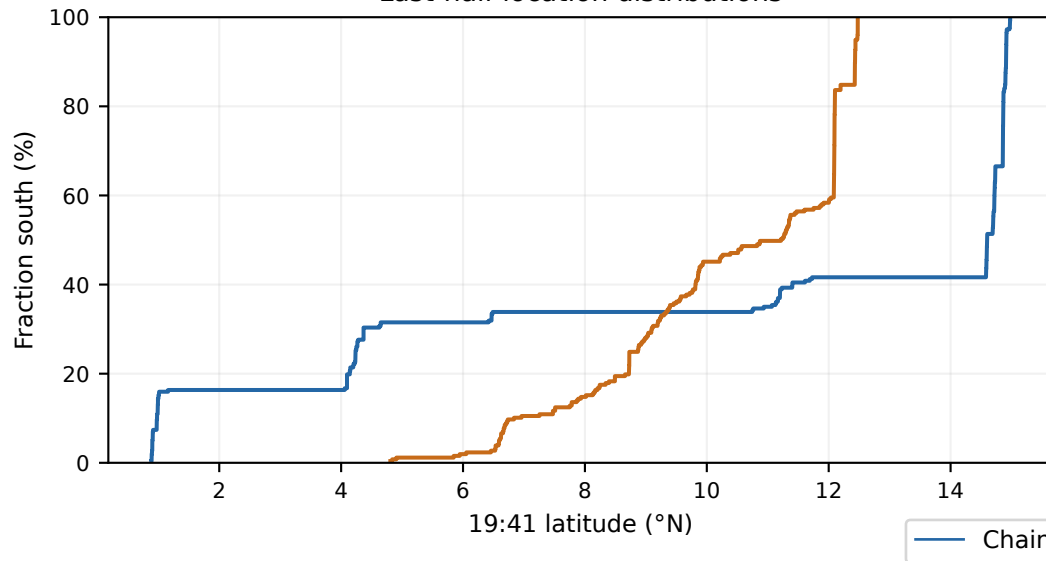
Full ancestor sampling: untempered continuation



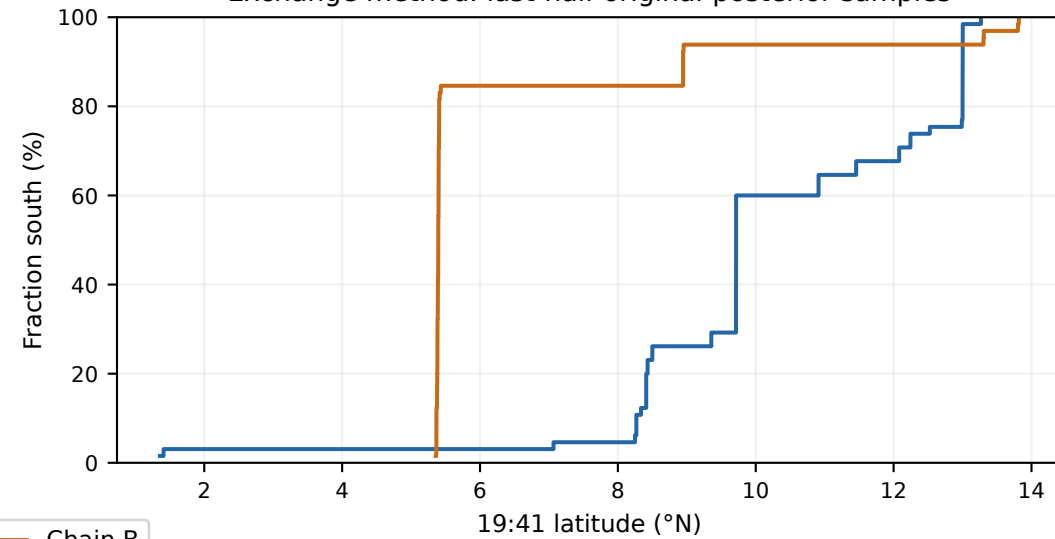
Ancestor sampling with likelihood-only exchanges



Last-half location distributions



Exchange method: last-half original-posterior samples



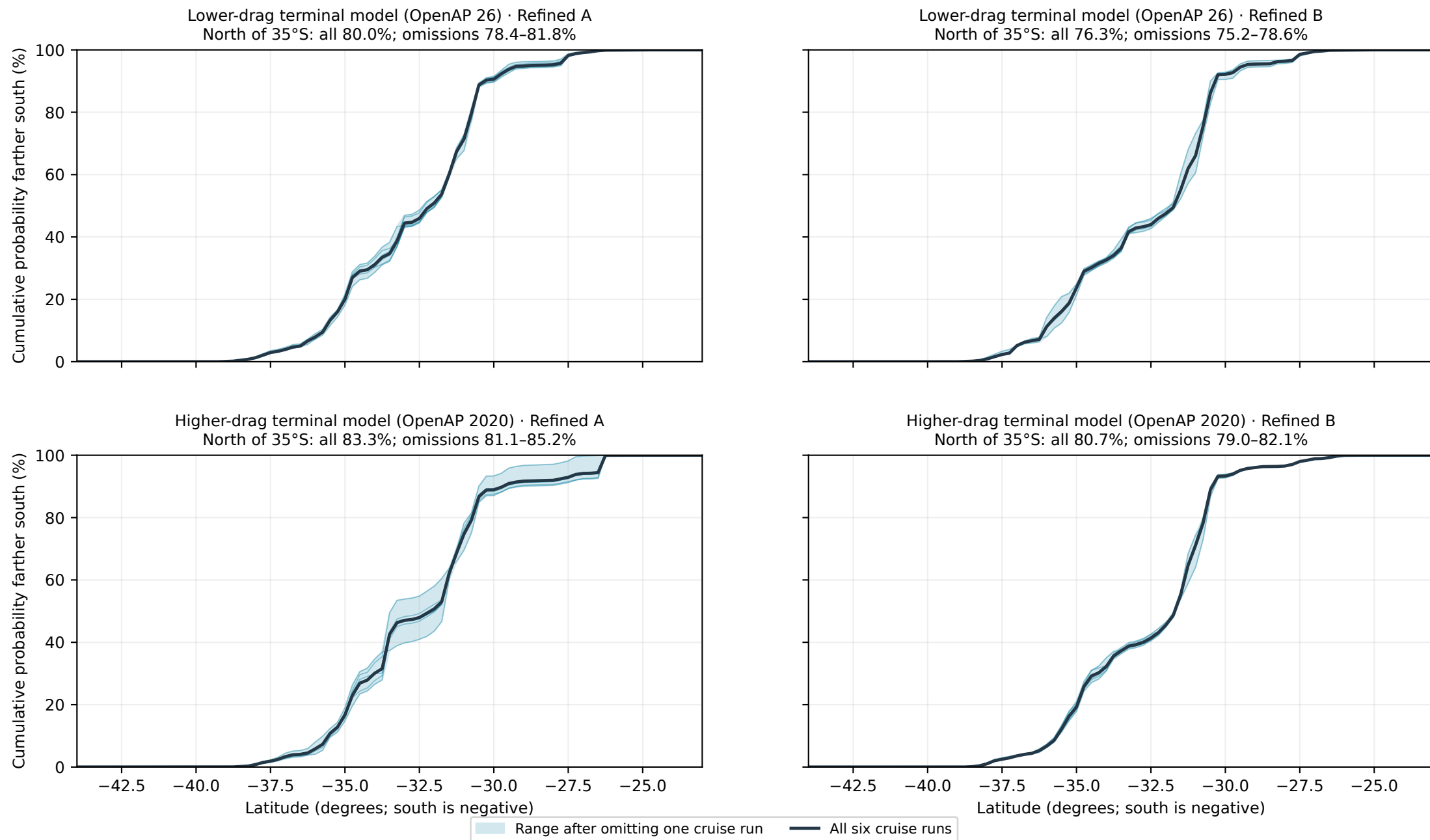
64 particles per sweep. Left: 512 additional sweeps after the pilot. Right: 128 sweeps $\times$ 5 fixed likelihood powers.

Last-half latitude CDF gaps: 58.4 pp without exchanges; 81.5 pp with exchanges. Different effort; short dependent samples.

Full suffix weights and shared BFO bias are recomputed. Flight priors stay unchanged; the final fuel condition remains at 00:11.

Dependent chain samples: movement is not convergence. Method: Lindsten, Jordan & Schön (2014), Algorithm 2; independent checks accompany the report.

How much does each original cruise run influence the mixed residual PDF?

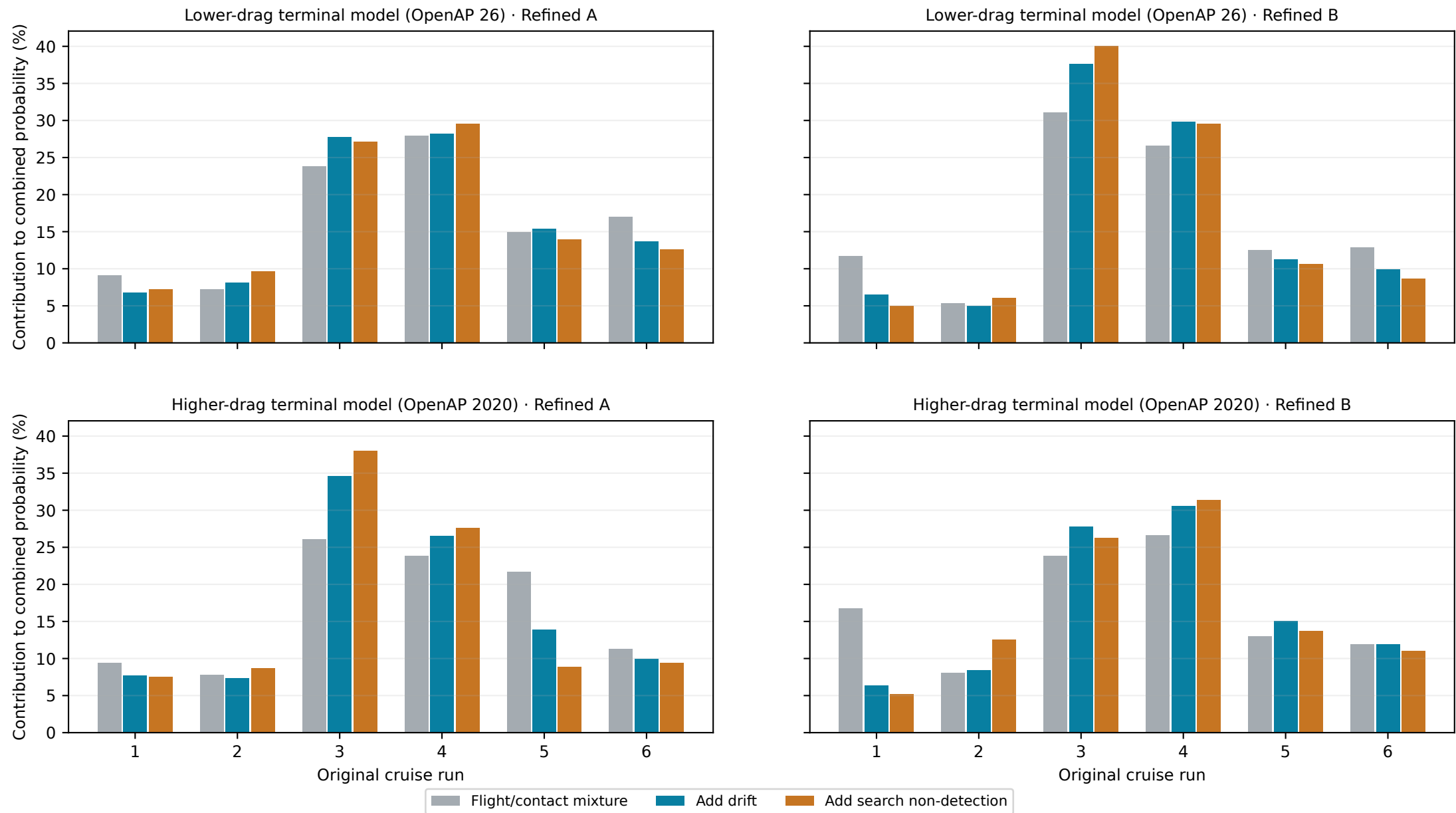


Nine recovered objects + ATSB 94% detection scenario. Each omission reassigns equal initial weights to the four BFO cases before evidence updating.

The band is source sensitivity, not a confidence interval. It includes each cruise population and its sampled terminal descendants.

No new trajectories; paths missed by every original run remain invisible. Cowling drift repeat A is shown; both repeats are in the data table.

Drift and search evidence redistribute weight among cruise populations



Cruise run seeds: 1: 37091631, 2: 37091640, 3: 37091711, 4: 37091712, 5: 37091713, 6: 37091714

Initial candidates: runs 1-2, 85,000 each; 3-4, 255,000 each; 5-6, 170,000 each. Sources have unequal original pooling weights.

Probability contribution is neither independent sample count nor evidence that one random seed is scientifically preferred.